IE509 Lab9 Pandas Assignment

Question 1

Along with this notebook is a data file named `daily_quotes.csv` which contains EOD OHLC/Volume data for a small number of equities over a 6 month period.

1. Load this data into a dataframe. Check the data types of each column. Check if you have datetime object for dates, floats for OHLC data, and integers for Volume.

2. Change the data types if necessary so that each column has the correct data type (`str`, `float`, `int`). Create a row index for data frame based on the `symbol` column.

3. Modify the dataframe so that it contains columns named and ordered as:

   - column 1: `symbol` (`str`)
   - column 2: `date` (`datetime`)
   - column 3: `open` (`float`)
   - column 4: `high` (`float`)
   - column 5: `low` (`float`)
   - column 6: `close` (`float`)
   - column 7: `volume` (`int`)

   (with `symbol` being used as the row index)

4. Create a dataframe which have the same structure as the original dataframe consisting of the row (or rows) containing the records with the highest volume for each specific symbol name (e.g., `AAPL`, `MSFT`, etc.)

5. Create a dataframe which have the same structure as the original dataframe consisting of the row (or rows) that have the largest high-low spread, for each specific symbol.

Question 2

1. Load the Dataset: First, load the dataset from the CSV file (`textual_data.csv`) using pandas.

2. Reshape data to have Person ID and Name as individual columns.

3. Split the Name column in the dataset into two separate columns: `First_Name` and `Last_Name`. Display the DataFrame after splitting.

4. Sort the DataFrame by `Last_Name` in alphabetical order and display the sorted DataFrame.

5. Add a new column to the DataFrame that calculates the length of each person's `First_Name`.

6. Filter the DataFrame to show only rows where the `Last_Name` starts with a specific letter (e.g., 'S').

7. Group the data by `Last_Name` and calculate the number of occurrences of each unique `Last_Name`.

8. Save the modified DataFrame (with split names) into a new CSV file named `"modified_textual_data.csv"`.